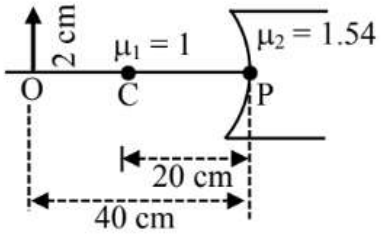
Q. 16 Refer the figure given below. μ_1 and μ_2 are refractive indices of air and lens material. The height of image will be _____ cm .



Option 1:
1

Option 2:
0.5

Option 3:
1.2

Option 4:
0.25

Correct Answer:
1

Solution:

$$\frac{\mu_2}{\mathrm{v}} - \frac{\mu_1}{\mu} = \frac{\mu_2 - \mu_1}{\mathrm{R}}$$

$$\frac{1.54}{\mathrm{v}} - \frac{1}{(-40)} = \frac{1.54 - 1}{-20}$$

$$\frac{1.54}{\mathrm{v}} = -\frac{1}{40} - \frac{0.54}{20}$$

$$\frac{1.54}{\mathrm{v}} = \frac{-2.08}{40}$$

$$\mathrm{v} = -29.61 \text{ cm}$$

$$\frac{\mathrm{h_i}}{\mathrm{h_0}} = \frac{\mu_1 \mathrm{v}}{\mu_2 \mathrm{u}} = \frac{1(-29.61)}{1.54(-40)} = 0.48$$

$$\mathrm{h_i} = 0.48 \times 2 = 0.96 \approx 1 \text{ cm}$$

Q. 17 For a certain metal, when monochromatic light of wavelength λ is incident, the stopping potential for photoelectrons is $3\ \mathrm{V_0}$. When the same metal is illuminated by light of wavelength 2λ, then the stopping potential becomes $\mathrm{V_0}$. The threshold wavelength for photoelectric emission for the given metal is $\alpha\lambda$.The value of α is

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Correct Answer:
33

Solution:

$$[Cr(H_2O)_6]Cl_3 + \underset{\text{excess}}{AgNO_3} \rightarrow \underset{\text{ppt}}{3AgCl(s)}$$

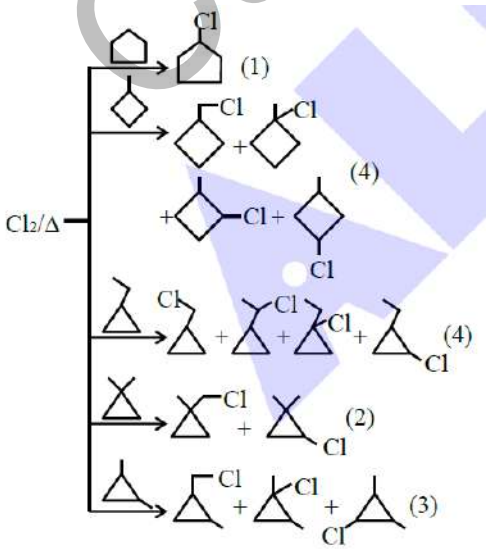
$$\frac{\text{no. of moles of complex reacted}}{\text{no. of moles of AgCl precipitated}} \times 100$$

$$= \frac{1}{3} \times 100 = 33.3$$

Q. 22 Consider the isomers of hydrocarbon with molecular formula C_5H_{10}. These isomers do not decolourise $KMnO_4$ solution. These isomers are subjected to chlorination with chlorine in presence of light to give monochloro compounds. The total number of monochloro compounds (structural isomers only) formed is _____ .

Correct Answer:
14

Solution:



1+4+4+2+3=14

Q. 23 One mole of an alkane (x) requires 8 mole oxygen for complete combustion. Sum of number of carbon and hydrogen atoms in the alkane (x) is _____ .

Correct Answer:
17

Solution:

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Q. 4

Let $A=\begin{bmatrix}1 & 2\\ 1 & \alpha\end{bmatrix}$ and $B=\begin{bmatrix}3 & 3\\ \beta & 2\end{bmatrix}$.

If $A^2-4A+I=O$ and $B^2-5B-6I=O$, then among the two statements :

(S1) : $[(\mathrm{B}-\mathrm{A})(\mathrm{B}+\mathrm{A})]^{\mathrm{T}}=\begin{bmatrix}13 & 15\\ 7 & 10\end{bmatrix}$

and

(S2) : $\det(\mathrm{adj}(\mathrm{A}+\mathrm{B}))=-5$,

Option 1:
only (S1) is correct

Option 2:
only (S2) is correct

Option 3:
both (S1) and (S2) are correct

Option 4:
both (S1) and (S2) are wrong

Correct Answer:
only (S2) is correct

Solution:

$\mathrm{A}=\begin{bmatrix}1 & 2\\ 1 & \alpha\end{bmatrix}$

$\mathrm{A}^2-4\,\mathrm{A}+\mathrm{I}=0$

trace of matrix

$\alpha+1=4$

$\alpha=3$

$\mathrm{B}=\begin{bmatrix}3 & 3\\ \beta & 2\end{bmatrix}$

$\mathrm{B}^2-5\,\mathrm{B}-6\mathrm{I}=0$

$\det(B)=6-3\beta=-6$

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$3\beta = 12$
$\beta = 4$
$\mathrm{B} - \mathrm{A} = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}; \mathrm{B} + \mathrm{A} = \begin{bmatrix} 4 & 5 \\ 5 & 5 \end{bmatrix}$
$(\mathrm{B} - \mathrm{A})(\mathrm{B} + \mathrm{A}) = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$
$\mathrm{adj}(\mathrm{A}) = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}, \mathrm{adjB} = \begin{bmatrix} 2 & -3 \\ -4 & 3 \end{bmatrix}$
$\mathrm{adj}(\mathrm{A} + \mathrm{B}) = \begin{bmatrix} 5 & -5 \\ -5 & 4 \end{bmatrix}$

Q. 5 Let A be the set of first 101 terms of an A.P., whose first term is 1 and the common difference is 5 and let B be the set of first 71 terms of an A.P., whose first term is 9 and the common difference is 7. Then the number of elements is $\mathrm{A} \cap \mathrm{B}$, which are divisible by 3 , is :

Option 1:
4

Option 2:
5

Option 3:
6

Option 4:
7

Correct Answer:
5

Solution:
AP_1 Set $\mathrm{A} = \{1, 6, 11, 16 \ldots\ldots\ldots\ldots 101 \text{ terms }\}$
AP_2 Set $\mathrm{B} = \{9, 16 \ldots\ldots\ldots\ldots 71 \text{ terms }\}$
$\mathrm{D} = \mathrm{L.C.M}\{\mathrm{d}_1,\ \mathrm{d}_2\} = 35$
1st Common term is 16

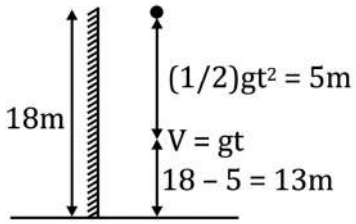
$16 + (\mathrm{n}-$
$\mathrm{n} \leq 14.8$
$\Rightarrow \mathrm{n} = 14$

$A \cap B = \{16, 51, 86, 121, 156, 191, 226, 261, 296,$ 331, 366,401,436,471\}
terms divisible by ' 3 '

$= \{51, 156, 261, 366, 471\}$

Ans. 5 terms

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At any time t after it is released velocity $\mathrm{V}=\mathrm{gt}$
Acceleration of the ball remains constant $=g$
$\therefore$ Both will be equal at $\mathrm{t}=1\mathrm{sec}$
$\therefore$ Position of ball from ground will be $(18-5)\mathrm{m}$

Q. 23 A transverse wave on a string is described by $\mathrm{y}=3\sin(36\mathrm{t}+0.018\mathrm{x}+\pi/4)$. Where x, y are in cm and t in seconds. The least distance between the two successive crests in the wave is _____ cm. (Nearest integer) $(\pi=3.14)$

Correct Answer:
349

Solution:


Least distance between successive crest $=\lambda$

$\mathrm{y}=\mathrm{A}\sin(\omega\mathrm{t}+\mathrm{kx}+\phi), \omega=2\pi\mathrm{f} \& \mathrm{k}=\dfrac{2\pi}{\lambda}$
$\&\mathrm{y}=\mathrm{A}\sin(36\mathrm{t}+0.018\mathrm{x}+\pi/4)$
$\Rightarrow \mathrm{k}=0.018\ \mathrm{cm}^{-1} \quad \Rightarrow \dfrac{2\pi}{\lambda}=0.018$
$\lambda=\dfrac{2\pi}{0.018}=\dfrac{2\times 3.14}{18\times 10^{-3}}\ \mathrm{cm}=348.88\ \mathrm{cm}$
$\approx 349\ \mathrm{cm}\ (\text{nearest integer})$

Q. 24 The charged particle moving in a uniform magnetic field of $(3\hat{i}+2\hat{j})\mathrm{T}$ has an acceleration $\left(4\hat{\mathrm{i}}-\frac{\mathrm{x}}{2}\hat{\mathrm{j}}\right)\mathrm{m/s}^2$. The value of x is

Correct Answer:
12

Solution:
$\overrightarrow{\mathrm{F}_\mathrm{B}} \propto (\overrightarrow{\mathrm{V}}\times\overrightarrow{\mathrm{B}}) \quad \Rightarrow \quad \overrightarrow{\mathrm{F}_\mathrm{B}} \perp \overrightarrow{\mathrm{B}}$
$\overrightarrow{\mathrm{a}} \perp \overrightarrow{\mathrm{B}} \rightarrow \overrightarrow{\mathrm{a}}\cdot\overrightarrow{\mathrm{B}}=0 \quad \Rightarrow$
$12-\mathrm{x}=0 \qquad x=12$

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Option 3:
1.0×10^{-5}

Option 4:
5×10^{-3}

Correct Answer:
1.0×10^{-5}

Solution:
$x_2 + y_2 \rightleftharpoons 2xy, \quad K' = \frac{1}{K_1} = \frac{1}{2.5 \times 10^5} = \frac{1}{25 \times 10^4}$

$\frac{1}{2}x_2 + \frac{1}{2}y_2 \rightleftharpoons xy, K'' = \left(\frac{1}{25 \times 10^4}\right)^{1/2} = \frac{1}{5 \times 10^2} \ldots\ldots$

$xy + \frac{1}{2}z_2 \rightleftharpoons xyz, K_2 = 5 \times 10^{-3}$

On adding equation (1) \& (2)

$\frac{1}{2}x_2 + \frac{1}{2}y_2 + \frac{1}{2}z_2 \rightleftharpoons xyz$

$K_3 = \frac{1}{5 \times 10^2} \times 5 \times 10^{-3} = 1 \times 10^{-5}$

Q. 6 Given at 298 K :

$\mathrm{E}^{\ominus}_{\mathrm{Fe}^{2+}/\mathrm{Fe}} = \mathrm{X\ Volt}$

$\mathrm{E}^{\ominus}_{\mathrm{Fe}^{3+}/\mathrm{Fe}} = \mathrm{Y\ Volt}$

The $\mathrm{E}^{\ominus}_{\mathrm{Fe}^{3+}/\mathrm{Fe}^{2+}}$ in Volt at 298 K is given by :

Option 1:
$2X - 3Y$

Option 2:
$3Y - 2X$

Option 3:
$3Y + 2X$

Option 4:
Y + X

Correct Answer:
$3Y - 2X$

Solution:

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$(i)\ Fe^{+2}(aq) + 2e^- \rightarrow Fe(s) \ldots\ldots\ldots \Delta G_1^\circ$

$(ii)\ Fe^{+3}(aq) + 3e^- \rightarrow Fe(s) \ldots\ldots\ldots \Delta G_2^\circ$

$(iii)\ Fe^{+3}(aq) + e^- \rightarrow Fe^{+2}(aq) \ldots\ldots\ldots \Delta G_3^\circ$

$iii = ii - i$

$-1 \times F \times E^\circ_{Fe^{3+}/Fe^{2+}} = -3 \times F \times y - (-2 \times F \times x)$

$E^\circ_{Fe^{3+}/Fe^{2+}} = 3y - 2x$

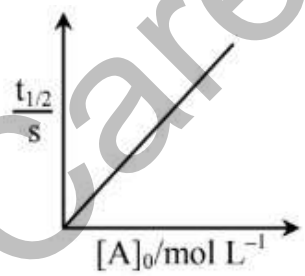
Option (2) is correct

Q. 7 Given below are two statements :

$R = 8.314\ J\ K^{-1}\ mol^{-1}$ and $1cal = 4.2\ J$

Statement I : When $Ea = 12.6 kcal/mol$, the room temperature rate constant is doubled by a $10^\circ C$ increase in temperature (298 K to 308 K)

Statement II : For a first order reaction $A \rightarrow B$,



Here $[A]_0$ is the initial concentration of A and $t_{1/2}$ is half-life of reaction.

In the light of the above statements, choose the correct answer from the option given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II are false

Option 4:
Statement I is false but Statement II are true

Correct Answer:
Statement I is true but Statement II are false

Solution:

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